

Independent machine verification of the fundamental-group computation in Wuebben’s exotic $S^2 \times S^2$

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Abstract

Wuebben (arXiv:2608.17267), building on Lidman–Piccirillo (arXiv:2505.14387), proves that a specified Lidman–Piccirillo piece V — a symplectic 4-manifold with the homology of $S^2 \times D^2$, built from a non-product genus-2 surface bundle over a once-punctured torus by two Luttinger surgeries — is simply connected; the symplectic double of V is then homeomorphic but not diffeomorphic to $S^2 \times S^2$, and Lidman–Piccirillo regluing yields an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$. The critical step, historically the failure point of claimed small exotic 4-manifolds, is the simple connectivity: the triviality of the filled fundamental groups for the paper’s permitted surgery parametrizations. We report a verification of that step independent of the paper’s coordinates, software, and author: the bundle is rebuilt as a simplicial complex from the paper’s marked-fiber data alone; the surgery tori, based meridians, and framed longitudes are extracted combinatorially; and the four theorem-critical filled presentations are proved trivial by complete confluent rewriting, with every computation emitting a replayable proof certificate validated by two independently written checkers. Four adjacent half-drift presentations pass the same certification as an auxiliary check. Wuebben’s committed scripts enter at exactly one point, parsed and never executed: after the independently reconstructed dictionary of conventions was frozen, selected entries were compared diagnostically against them, and no verified conclusion consumes script-derived data. The paper’s framing lemma is machine-checked with its analytic normalizations rebuilt constructively and chart independence proved between the paper’s displayed charts. Stated carefully: the simple-connectivity proof appears complete relative to explicitly named standard topology theorems, with no known project-specific gap. The theorem is Wuebben’s; this note claims only the verification. All certificates, code, and provenance are public in a pinned release.

1 Introduction

Claimed constructions of small exotic 4-manifolds have repeatedly failed in one place: the fundamental-group computation. The manifolds of [1] rest on a specified Lidman–Piccirillo piece V , obtained by two Luttinger surgeries [5, 3] on Lagrangian tori in a symplectic non-product genus-2 surface bundle R over a once-punctured torus, following the framework of [2]. Exoticness of the double of V rests on V being simply connected: the filled fundamental groups of the paper’s permitted surgery parametrizations — the four sign choices of the double surgery in the paper’s principal coordinate system — must be trivial. Four adjacent

“half-drift” presentations of the same fillings serve as re-indexing checks; they carry no independent logical load.

This note reports an independent verification of that step, together with a machine audit of the framing lemma that connects the group theory to the symplectic geometry. Independence here is concrete: nothing in the verification reuses the paper’s coordinates, code, or intermediate data. The geometry was rebuilt from the paper’s stated marked-fiber data alone, and the paper’s displayed relations, sign tables, and Table 1 filling words were then *compared against* the rebuild rather than assumed. The downstream deductions — from simple connectivity to the homeomorphism classifications and the exoticness conclusions — are Wuebben’s; this note verifies the simple-connectivity input and its geometric framing bridge, and a hypothesis audit of the downstream deductions is recorded in the repository.

Two disclosures frame everything that follows. First, the theorem is Wuebben’s; this note asserts nothing about priority or authorship of the mathematics and claims only the verification. Second, the verification was carried out by generative-AI systems under the author’s direction, with commit-level provenance recorded in the repository’s ledger; mathematical responsibility rests with the human author. No claim below depends on trusting a model: every asserted computation ships with a replayable certificate.

Main verification result. *From a fresh checkout of release v1.2 of the verification repository (Section 6): (i) the four theorem-critical filled presentations — the $n=0$ sign pairs of the paper’s double surgery, imposed with independently certified based meridians and framed longitudes — carry complete confluent rewriting certificates proving each group trivial, and the certificates replay and are accepted by two independently written checkers; (ii) the four adjacent $n=1$ half-drift presentations carry the same certificates, as an auxiliary check; (iii) the external inputs are the named standard theorems of Section 6, which this note does not reprove; (iv) every certificate, transcript, and the provenance ledger are contained in that pinned release.*

2 The independent model and the interpretation dictionary

The genus-2 fiber is realized as a regular neighborhood of the paper’s five-chain of curves: five plumbed annular bands and two cone discs, with the hyperelliptic-type involution φ_0 realized simplicially. Its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α , T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

The identification with the paper’s marked bundle is by literal based monodromy action: the based mapping classes of both monodromies are certified as equalities of vertex paths, transported by proof-producing Tietze chains that replay step by step. The identification is completed by an explicit marked mapping-cylinder clutching over the base graph: the relative monodromy comparisons are exact on the marked collars, so the identification is a constructed fiberwise homeomorphism into the paper’s bundle rather than an appeal to the classification of surface bundles. Everything the surgery argument needs is transported through that map into the paper’s already-smooth manifold, where the smooth and symplectic arguments run; no smoothing of the triangulation is ever chosen.

Because the machine certifies properties of a model, the model itself is the residual risk.

Five structurally different implementations reconstruct it independently — among them an independent peripheral extractor built only from the bundle, an alternative triangulation with its own based-monodromy certificate, and a second marked-fiber realization built directly from the abstract ribbon graph — and they are structured so that the classes of translation error they can detect — subdivision artifacts, monodromy transcription, ordering, sign, and orientation conventions — would each be caught by at least one independent reconstruction. What redundancy cannot exclude, a shared misreading of the paper’s conventions, is addressed by a bound interpretation dictionary: every convention read from the paper carries the certified computation that would fail under a rival reading, with selected entries compared diagnostically — parsed, never executed — against Wuebben’s committed scripts after the dictionary was frozen, so that no verified conclusion consumes script-derived data, and the remaining transcription-level entries audited by a quarantined extractor against the hash-pinned immutable Lidman–Piccirillo v1 arXiv source [2], in both its prose and its vector-figure layers. The one interpretation remaining at that boundary is the ordinary convention that dashed arcs in a handle picture depict hidden projection rather than extra crossings. The full inventory of the redundant implementations and the dictionary is in the repository.

3 Peripheral data and the framing identification

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper’s own basing whiskers. The paper’s sign-correction package (its §7.4 fix) is reproduced independently, and a combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper’s corrected convention requires. Both surgery tori are certified locally flat at every one of their simplex links, each classified against the correct unknotted model, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked.

Luttinger surgery is performed with respect to the Lagrangian framing; the paper’s Lemma 8.2 identifies it with the fibered framing carried by the combinatorial longitudes above. This lemma carries an exact logical load: if the identification failed by j meridians, the certificates of Section 4 would remain correct for the product-framed presentations they literally describe, but would no longer identify those presentations with the paper’s surgeries. The lemma’s displayed Weinstein-chart, seam, and constant-momentum algebra are machine-checked symbolically, and its inline analytic normalizations have been rebuilt constructively rather than cited: the relative Moser flow is produced as the explicit monotone inverse of a cumulative coordinate, so no ODE existence theorem is invoked, and the equivariant step defines the Moser field directly on the double cover with certified projection and deck invariance, so no covering-lift theorem is invoked. Chart independence is proved between the paper’s displayed charts by two routes, with the one symplectic move that could introduce meridian components — the closed-1-form shift — excluded by the zero-section condition; [3] (§2.1 and Proposition 2.2) stands as corroboration.

Independently of the lemma, a finite 100-case scan over selected small framing shifts refills the complement with meridian-shifted longitudes and runs the identical certification pipeline on each counterfactual: 82 of the 100 cases are certified trivial, no nontriviality witness was found, and 18 remain undecided. Every shift touching only T_β is certified trivial in both families,

and each undecided case is perfect, admits no epimorphism onto the six smallest nonabelian simple groups, and resists coset enumeration to a ceiling of 12,499,998 cosets. These are counterfactual groups: the paper’s own fillings are the certified controls, and the undecided cases have no bearing on the proof unless the framing argument above first fails. The full table and methods are in the repository.

4 Certified triviality of the four theorem-critical groups

Filling the triangulation-derived complement presentation (4 generators, 95 relators) with the certified coherent peripheral pairs yields, for all four $n=0$ sign pairs, complete confluent rewriting systems in which every generator reduces to the identity.

system	signs	derivation records
$n=0$	$(-, -)$	1123
$n=0$	$(-, +)$	1075
$n=0$	$(+, -)$	804
$n=0$	$(+, +)$	1504

Derivation-DAG certificates for the four theorem-critical filled groups; each verifies TRIVIAL from a fresh clone of the repository.

The four adjacent $n=1$ half-drift presentations pass the identical certification as an auxiliary check; their certificates sit beside these in the same manifest.

5 Certificate verification and reproducibility

Knuth–Bendix completion (KB MAG [11], run under GAP [12]) is used only as an *untrusted certificate generator*: the exported derivation DAGs are validated by a small independent checker that re-verifies every input-relator axiom, inverse axiom, critical overlap, rewrite trace, and final generator-to-identity rule, and binds the input presentation by SHA-256. The checker neither imports nor runs KB MAG. A second implementation, written independently in Ruby, re-verifies all retained derivation records against the same hash-bound DAGs and accepts, importing neither the first checker nor any project group code. All artifacts — the certificates, the exported rewriting systems, and a hash manifest — replay from a clean checkout, and the stored rewriting systems reload and re-verify under GAP/KB MAG independently of the certificate path.

The verification was AI-assisted throughout, by generative-AI systems working under the author’s direction; the repository’s provenance ledger records which system produced which artifact, and preserves the corrections that adversarial cross-model review produced rather than rewriting them. Mathematical responsibility rests with the human author. The design principle throughout is that certificates are checked, not trusted: search and completion tools are treated as untrusted generators, and every computational claim rests on a replayable artifact validated by small independent code.

6 Trust boundary and data availability

The named standard inputs, applied with checked hypotheses: Freedman’s simply connected classification [6] with the Hambleton–Kreck finite-cyclic classification [7, 8]; symplectic Kodaira dimension and minimality, via the audited Luttinger-surgery constructions [3, 9]; the symplectic Thom theorem [10]; the Rokhlin/Arf input of the slicing argument, as used in [2]; the Ozsváth–Szabó / Lidman–Piccirillo Floer machinery [2]; and the Kerékjártó periodic-disk involution theorem [4] behind the disk extension of the paper’s Lemma 7.1. Alongside these sit elementary inputs of PL topology retained as standard without a specific citation: the ribbon-thickening and bistellar-trace interpretations that read the paper’s figures into complexes, the classification of surfaces, the collapsible-3-ball and cyclic-knot-unknot criteria behind the local-flatness certificates, and simplicial fundamental-group presentation with Tietze theory. This note reproves none of these. The ribbon-interpretation input is narrower than the phrase suggests: the certified classification of equivariant ribbon rotations pins the implication from the paper’s stated marked five-chain data to the equivariant normal form, and the stated data themselves are audited in the original Lidman–Piccirillo source, so what this boundary retains is the thickening of that data into complexes together with the ordinary dashed-arc hidden-projection convention of the audited figure. The accurate one-sentence status is: *the simple-connectivity proof appears complete relative to explicitly named standard topology theorems, with no known project-specific gap* — a statement about verification depth, offered to facilitate expert review, not to replace it.

All code, certificates, run transcripts, referee packets, and the provenance ledger are contained, with clean-checkout replay instructions, in the tagged release v1.2 of the repository: <https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v1.2>; the certificate manifest binds every artifact by SHA-256. The paper’s own code is at <https://github.com/bwuebben/exotic-s2xs2>; nothing in the verification derives from it except where explicitly cited.

References

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